Statistical modeling for causal inference

Cornell STSCI / INFO / ILRST 3900
Fall 2024
causal3900.github.io

Oct 20 2026

Learning goals for today

At the end of class, you will be able to

- Estimate average causal effects via g-formula with a statistical model for the outcome and treatment
- Reason about the bias variance tradeoff when selecting a statistical model

Standardization

Under consistency and conditional exchangeability and consistency:

$$\mathsf{E}(Y \mid A = 1, L = \ell) = \mathsf{E}(Y^{a=1} \mid A = 1, L = \ell) = \mathsf{E}(Y^{a=1} \mid L = \ell)$$

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The ATE can be identified via:

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&\mathsf{E}(Y^{a=1}) - \mathsf{E}(Y^{a=0}) \\
&\quad = \sum_{\ell} \underbrace{\Big[\mathsf{E}(Y^{a=1} \mid L = \ell) - E(Y^{a=0} \mid L = \ell)\Big]}_{\text{Conditional ATE for } \ell} \underbrace{Pr(L = \ell)}_{\text{Prop of group } \ell} \\
&\quad = \underbrace{\sum_{\ell} \Big[\mathsf{E}(Y^{a=1} \mid L = \ell) Pr(L = \ell)\Big]}_{\mathsf{E}(Y^{a=1})} - \underbrace{\sum_{\ell} \Big[\mathsf{E}(Y^{a=0} \mid L = \ell) Pr(L = \ell)\Big]}_{\mathsf{E}(Y^{a=0})}
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Aggregate the average over sub-groups to get the overall average

$$\mathsf{E}(Y^a) = \sum_{\ell} \underbrace{\mathsf{E}(Y^a \mid L = \ell)}_{\text{Avg of sub-group}} \quad \times \quad \underbrace{Pr(L = \ell)}_{\text{Prob of sub-group}}$$

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We want to estimate:

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Given data, we can

- Estimate $Pr(L = \ell)$ use observed proportion of people with $L_i = \ell$

$$\widehat{Pr}(L = \ell)$$

- Estimate $\mathrm{E}(Y^{a=1} \mid L = \ell)$, use observed average outcome for people with $A_i = 1$, $L_i = \ell$

$$\hat{\mathrm{E}}(Y^{a=1} \mid L = \ell) = \hat{\mathrm{E}}(Y \mid A = 1, L = \ell)$$

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Plug-in estimates for each quantity

$$\hat{\mathrm{E}}(Y^a) = \sum_\ell \underbrace{\hat{\mathrm{E}}(Y^a \mid L = \ell)}_{\text{Observed avg of sub-group}} \quad \times \quad \underbrace{\widehat{Pr}(L = \ell)}_{\text{Observed prob of sub-group}}$$

Estimating from data

Two small tweaks

- Estimate $Pr(L = \ell)$ use observed proportion of people with $L_i = \ell$

$$\widehat{Pr}(L = \ell)$$

- Rewrite as an average across individuals (instead of groups)

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- Fit a model to estimate

$$\hat{\mathrm{E}}(Y^{a=1} \mid L = \ell) = \hat{\mathrm{E}}(Y \mid A = 1, L = \ell)$$

Standardization

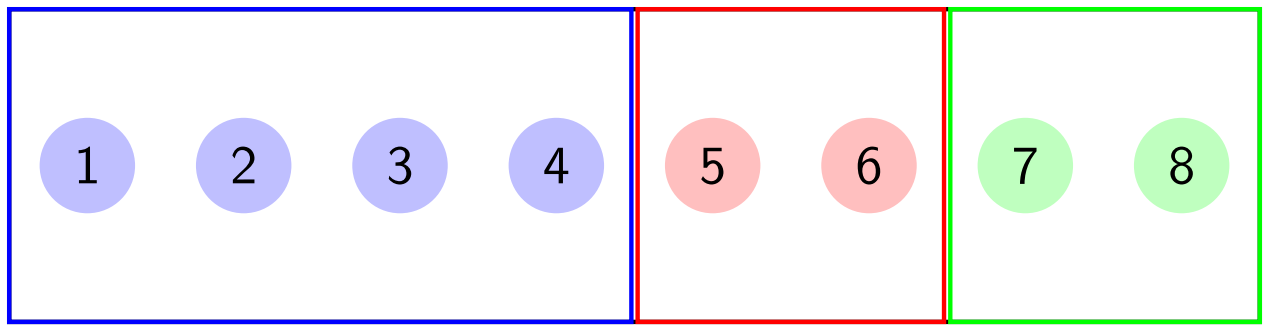
Aggregate the average over sub-groups to get the overall average

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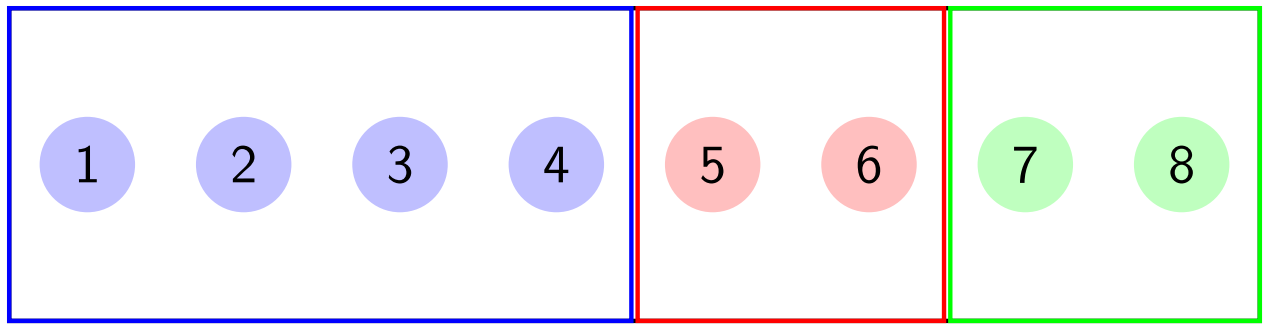


$$\hat{\mathrm{E}}(Y^a) = \hat{\mathrm{E}}(Y^a \mid L = \bullet) \times \frac{4}{8} + \hat{\mathrm{E}}(Y^a \mid L = \bullet) \times \frac{2}{8} + \hat{\mathrm{E}}(Y^a \mid L = \bullet) \times \frac{2}{8}$$

Standardization

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Calculate the same quantity but sum over individuals

$$\begin{aligned}\hat{\mathrm{E}}(Y^a) = \Big(&\hat{\mathrm{E}}(Y^a \mid L = \text{color}_1) + \hat{\mathrm{E}}(Y^a \mid L = \text{color}_2) + \hat{\mathrm{E}}(Y^a \mid L = \text{color}_3) \\ &+ \hat{\mathrm{E}}(Y^a \mid L = \text{color}_4) + \hat{\mathrm{E}}(Y^a \mid L = \text{color}_5) + \hat{\mathrm{E}}(Y^a \mid L = \text{color}_6) \\ &+ \hat{\mathrm{E}}(Y^a \mid L = \text{color}_7) + \hat{\mathrm{E}}(Y^a \mid L = \text{color}_8)\Big)/8\end{aligned}$$

Summing over individuals

Aggregate the average over sub-groups to get the overall average

$$\hat{\mathrm{E}}(Y^a) = \sum_{\ell} \underbrace{\hat{\mathrm{E}}(Y^a \mid L = \ell)}_{\text{Avg of sub-group}} \times \underbrace{\hat{Pr}(L = \ell)}_{\text{Prob of sub-group}}$$

$$= \frac{1}{n} \sum_{i} \underbrace{\hat{\mathrm{E}}(Y^a \mid L = \ell_i)}_{\text{Avg of sub-group for unit i}}$$

Estimating from data

Two small tweaks

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$$\widehat{Pr}(L = \ell)$$

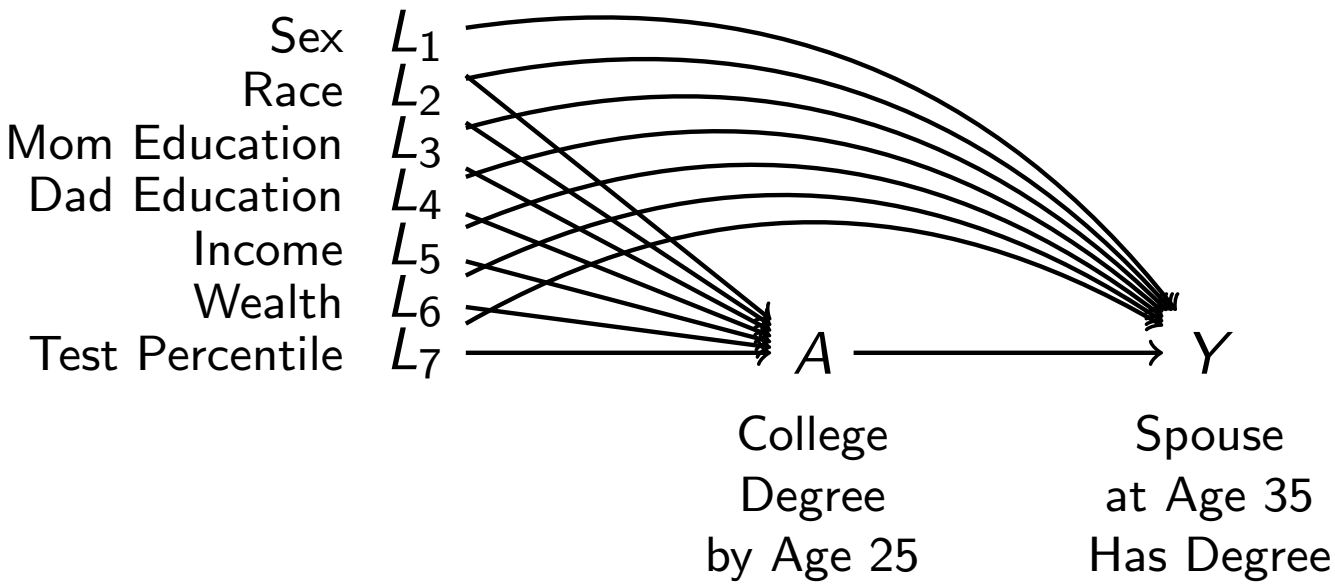
- Rewrite as an average across individuals (instead of groups)
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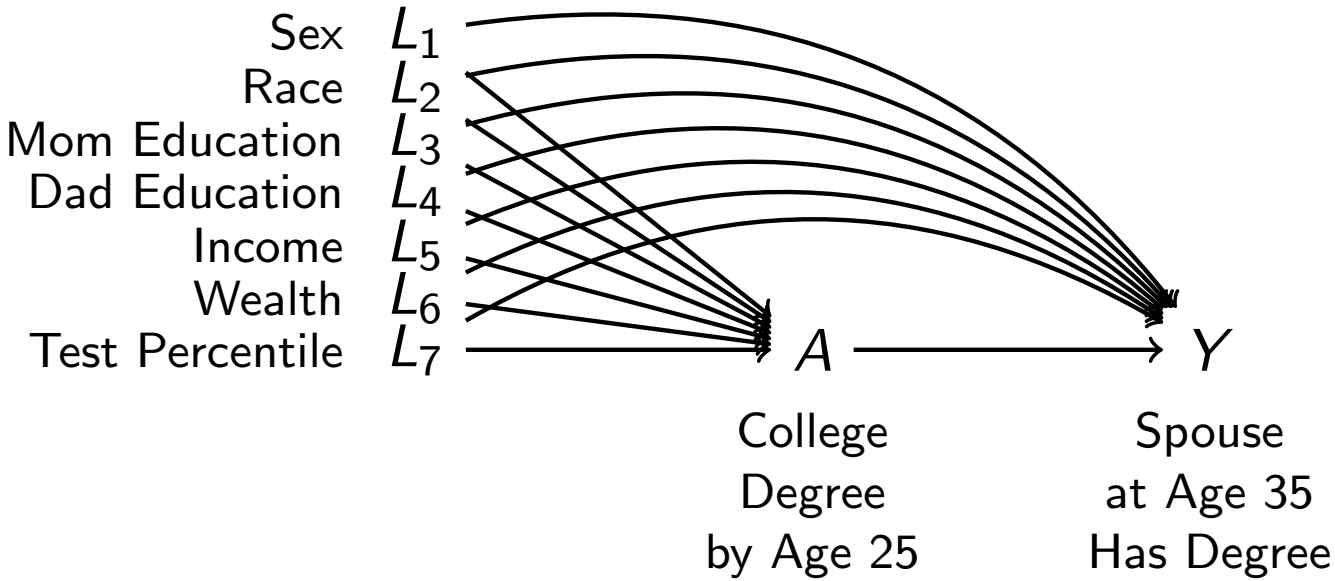
- Fit a model to estimate

$$\hat{\mathrm{E}}(Y^{a=1} \mid L = \ell) = \hat{\mathrm{E}}(Y \mid A = 1, L = \ell)$$

Nonparametric estimation breaks down



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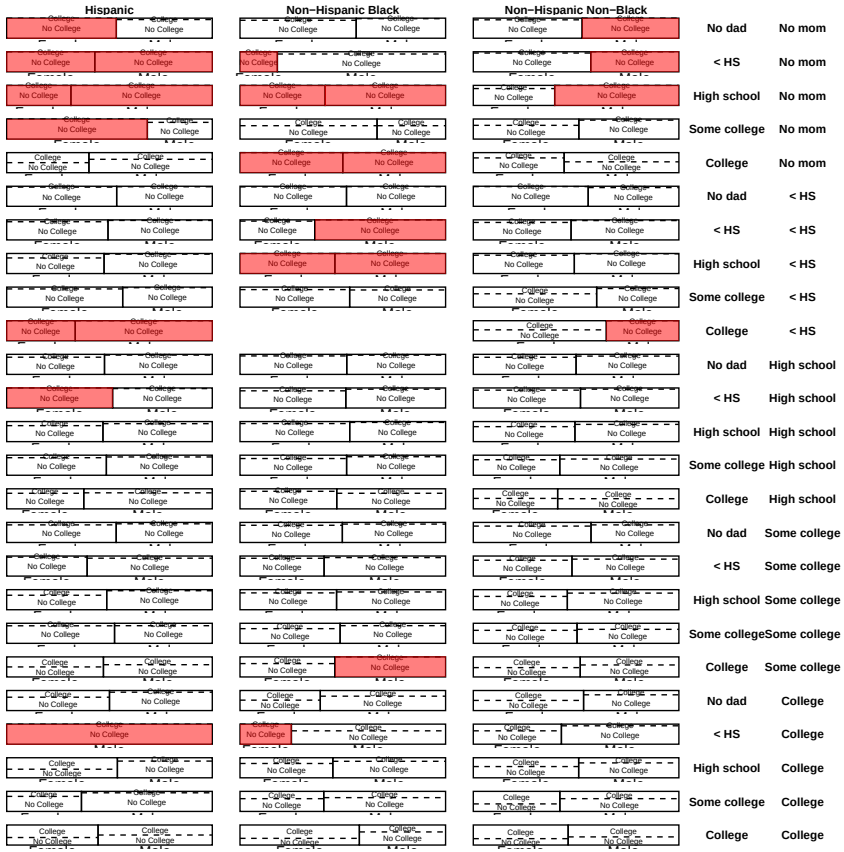
We need both:

$$\hat{\mathrm{E}}(Y \mid A = 1, L = \ell_i)$$

$$\hat{\mathrm{E}}(Y \mid A = 0, L = \ell_i)$$

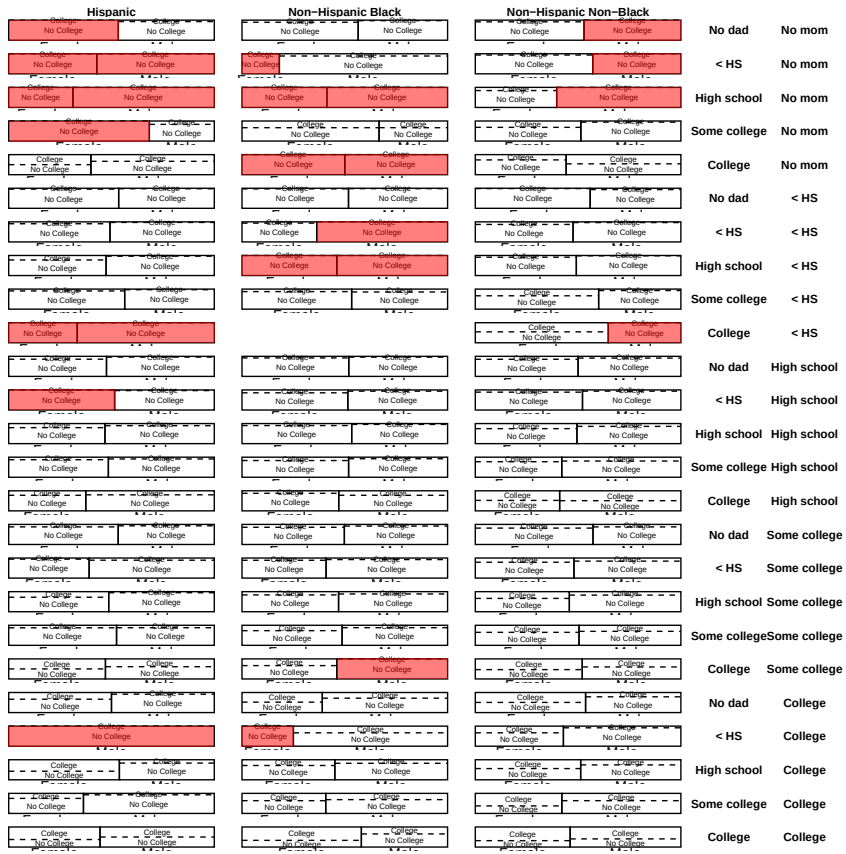
Nonparametric estimation breaks down

Can't estimate $\hat{\mathrm{E}}(Y \mid L = \ell_i, A = a)$ by taking an average for every sub-group



Nonparametric estimation breaks down

Can't estimate $\hat{\mathrm{E}}(Y \mid L = \ell_i, A = a)$ by taking an average for every sub-group



We instead use a statistical model to estimate

$$\hat{\mathrm{E}}(Y \mid L = \ell_i, A = a)$$

Parametric estimation: Outcome model

Model the conditional expectation of Y given L and A

- Linear regression

$$\hat{\mathrm{E}}(Y \mid L, A) = \hat{\alpha} + L'\hat{\gamma} + A\hat{\beta}$$

Parametric estimation: Outcome model

Model the conditional expectation of Y given L and A

- Linear regression

$$\hat{\mathrm{E}}(Y \mid L, A) = \hat{\alpha} + L'\hat{\gamma} + A\hat{\beta}$$

$$\hat{\mathrm{E}}(Y_i \mid \text{Test Score}_i, A_i) = .2 + .003 \times \text{Test Score}_i + .2 \times A_i$$

- If $\text{Test Score}_i = 80$ and $A_i = 1$ then

$$\hat{\mathrm{E}}(Y \mid \text{Test Score}, A) = .2 + .003 \times (80) + .2(1) = .64$$

- If $\text{Test Score}_i = 62$ and $A_i = 0$ then

$$\hat{\mathrm{E}}(Y \mid \text{Test Score}, A) = .2 + .003 \times (62) + .2(0) = 0.386$$

G-formula

Causal assumptions



Learn a model to predict Y given L and A

$$\hat{\mathrm{E}}(Y \mid L, A) = \hat{\alpha} + L'\hat{\gamma} + A\hat{\beta}$$

G-formula

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Learn a model to predict Y given L and A

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Standardization estimator

$$\hat{\mathrm{E}}(Y^a) = \frac{1}{n}\sum_i \hat{\mathrm{E}}(Y \mid L = \ell_i, A = a)$$

G-formula

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Learn a model to predict Y given L and A

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Standardization estimator

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For every unit i,

- Set the treatment value to a
- Predict the outcome

Then average over all units

The parametric g-formula: Connection to $\hat{\beta}$

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Estimator for the effect $\mathrm{E}(Y^1) - \mathrm{E}(Y^0)$:

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$$\hat{\mathsf{E}}(Y^1) - \hat{\mathsf{E}}(Y^0) = \left(\frac{1}{n} \sum_{i=1}^{n} \left(\hat{\alpha} + \hat{\gamma} \ell_i + \hat{\beta} \times 1 \right) \right) - \left(\frac{1}{n} \sum_{i=1}^{n} \left(\hat{\alpha} + \hat{\gamma} \ell_i + \hat{\beta} \times 0 \right) \right)$$

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With OLS, the parametric g-formula collapses on the coefficient.

The parametric g-formula allows for more complex models

As long as we have a model for predicting $E(Y \mid L, A)$, we can apply the g-formula

- Linear models with interaction terms
- Other types of regression: logistic regression, poisson regression, etc
- Machine learning models
  - Deep Neural Networks
  - Random forests
  - etc

Bias Variance trade-off

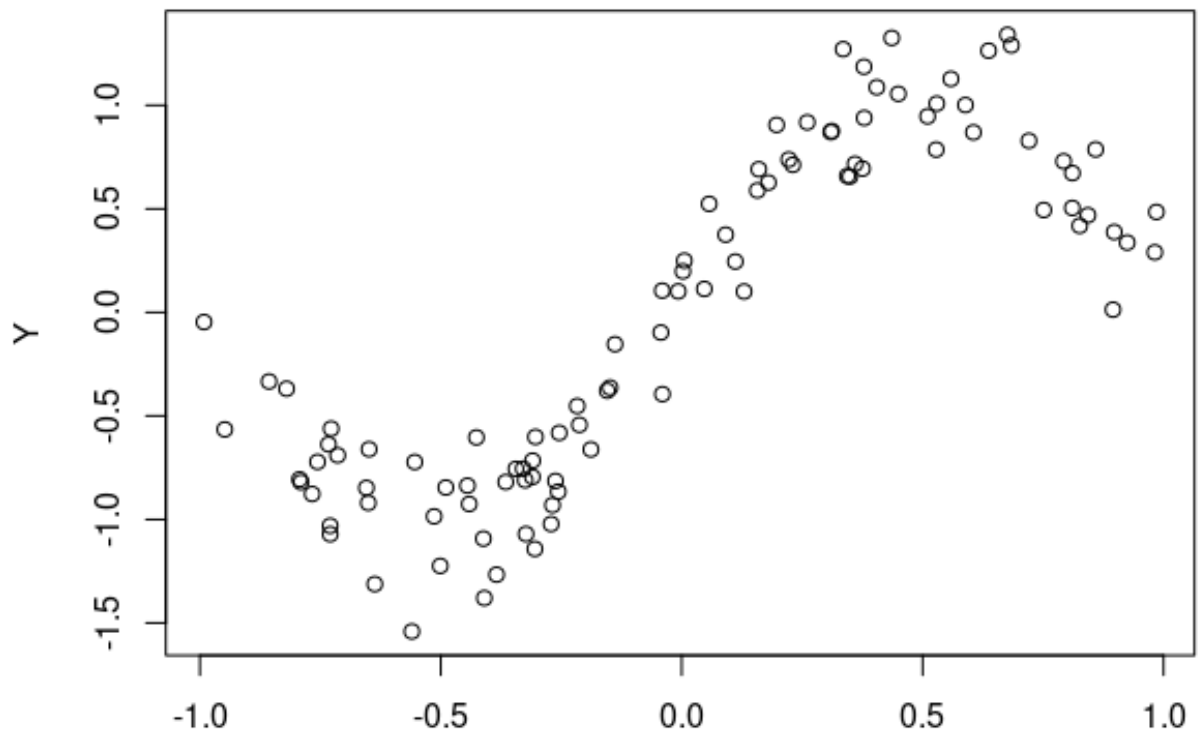
In statistics, the bias variance trade off is a fundamental constraint

- **Bias**: The functions we may estimate are not complex enough to capture the "true relationship"
- **Variance**: The model we are fitting is too complex so our estimated parameters change a lot from sample to sample

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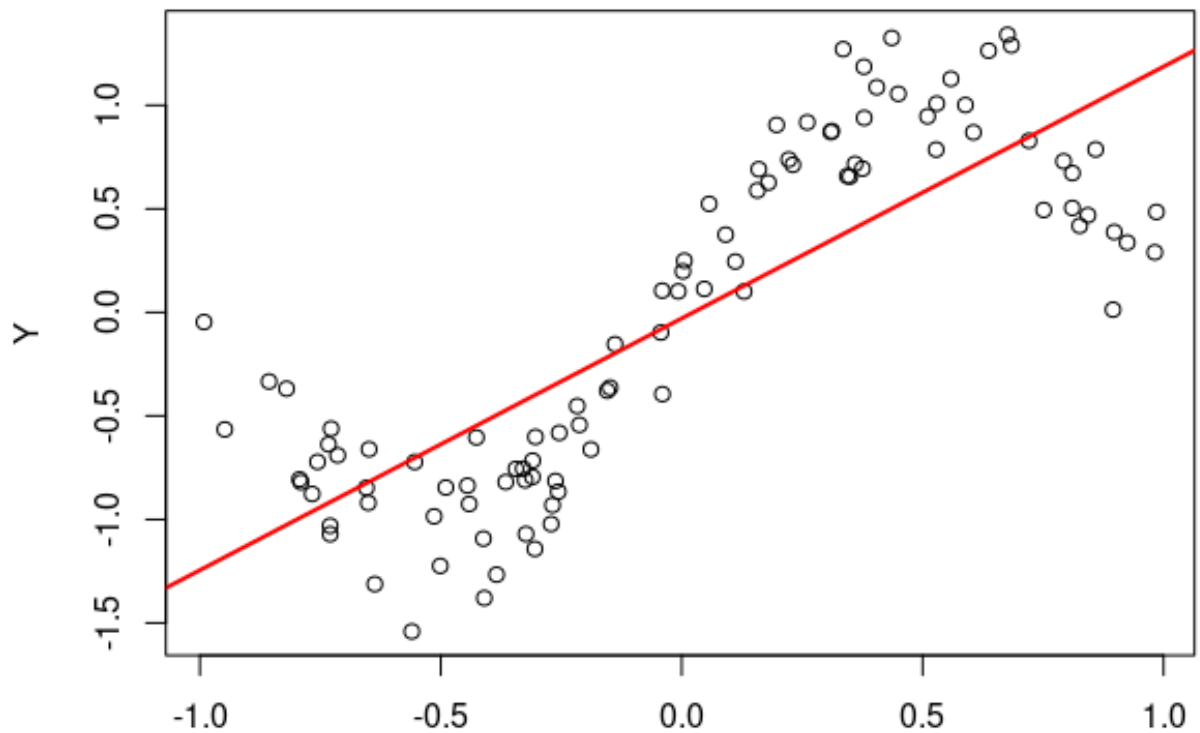
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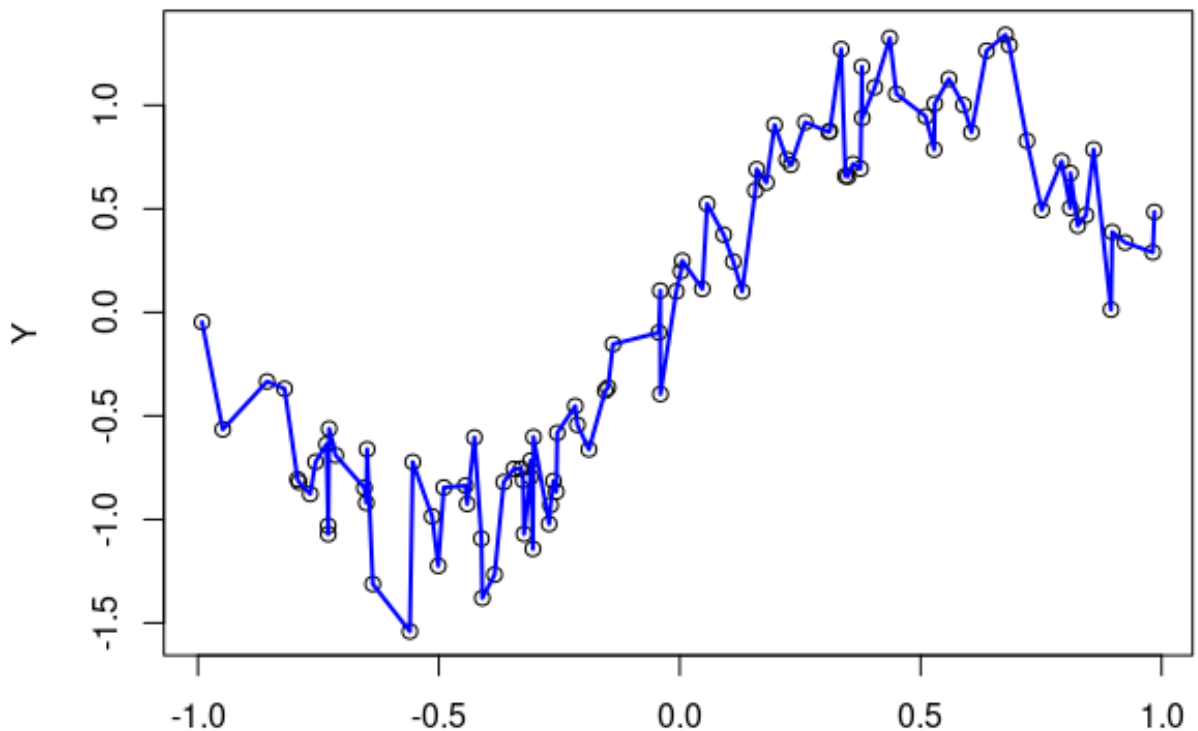
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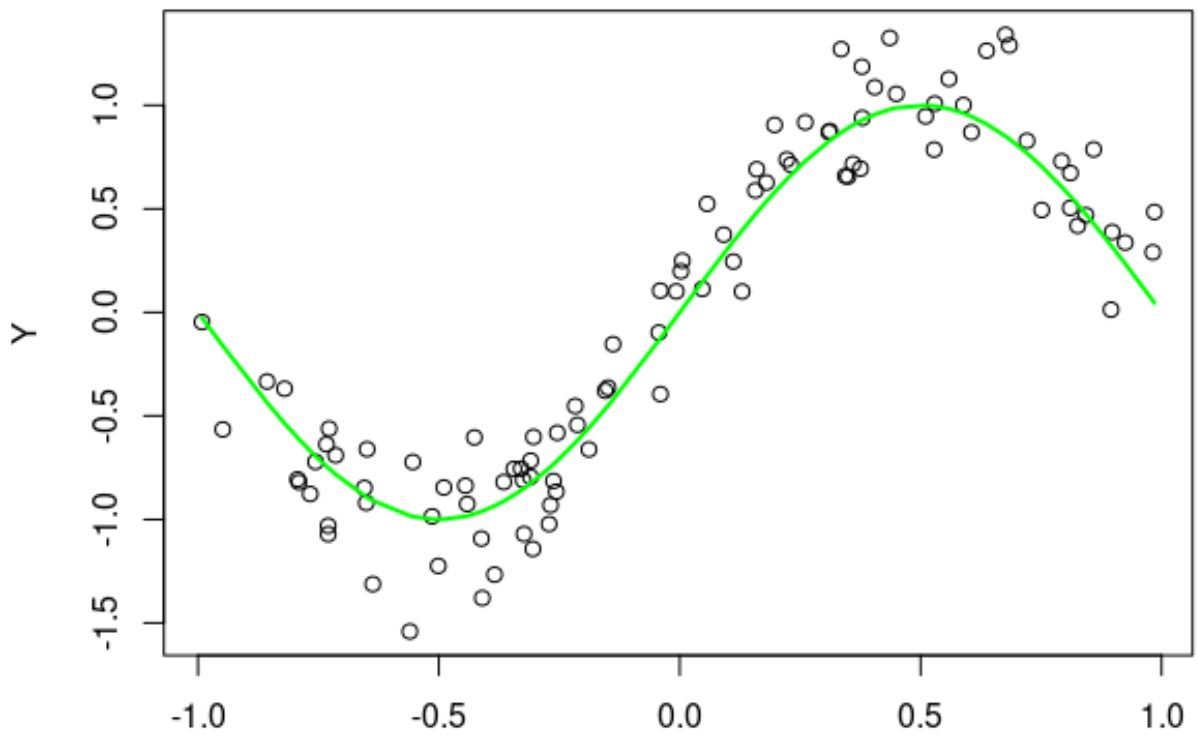
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Bias Variance trade-off

Bias and variance in making cakes:



Figure: High Bias, low variance



Figure: Low bias, High variance

Bias and variance in choosing conditional expectation model

- Linear model: 1 parameter per covariate (probably high bias)
- Observed average: 2^p means to estimate for p binary covariates (probably high variance)
- Other methods are typically somewhere in between
- Larger sample allows for more complex models

Bias and variance in choosing causal model

- Is a a DAG ever "truly correct"?
- Can always add more confounders
- Would the bias from the confounders you could add substantially change your claim?
- Including additional confounders makes estimation more difficult

Parametric g-formula: Outcome model recap

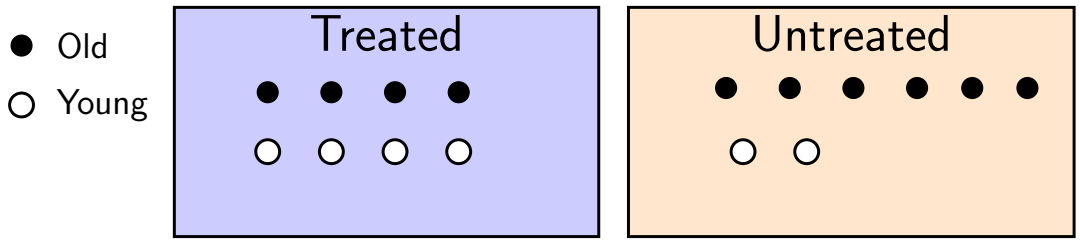


1. Estimate the outcome mean $\mathrm{E}(Y \mid A, L)$ with some model
2. Change everyone's treatment to the value of interest
3. Predict for everyone
4. Take the average

$$\hat{\mathrm{E}}(Y^a) = \frac{1}{n} \sum_{i=1}^{n} \hat{\mathrm{E}}(Y \mid L = \ell_i, A = a)$$

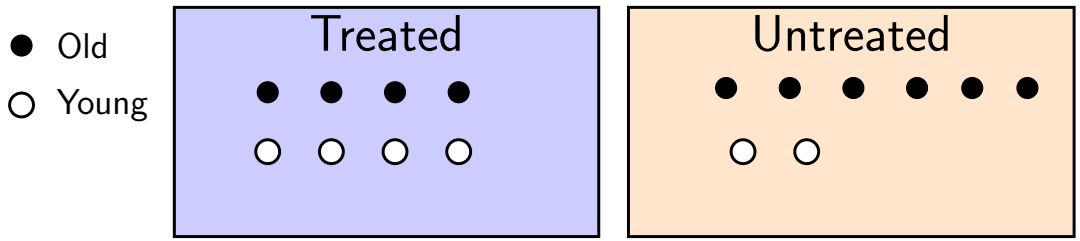
Inverse probability weighting

Observe: A conditionally randomized experiment:



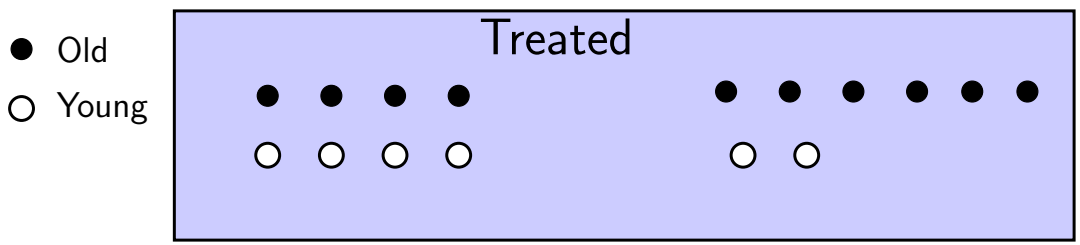
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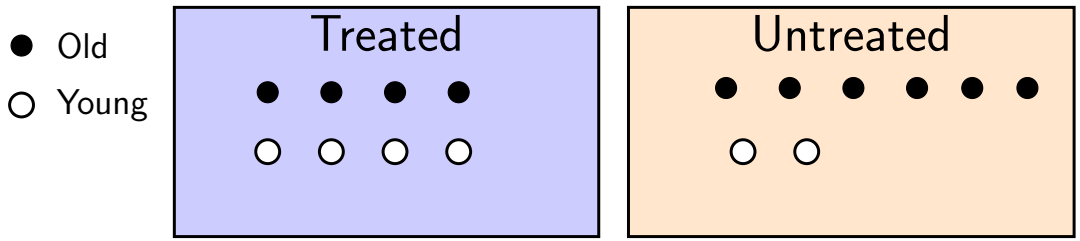
In order to get the ATE $= E(Y^{a=1}) - E(Y^{a=0})$

Want: a world where everyone is treated



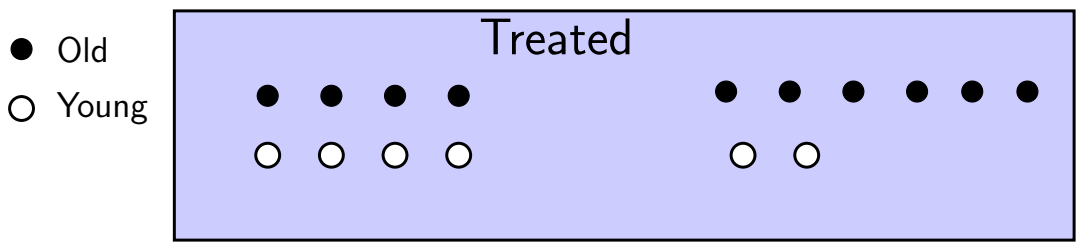
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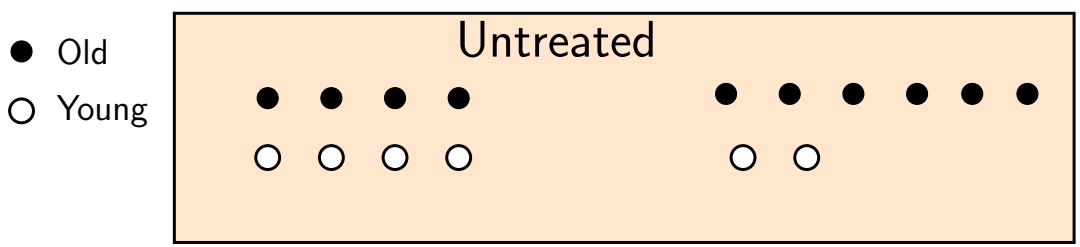


In order to get the ATE $= E(Y^{a=1}) - E(Y^{a=0})$

Want: a world where everyone is treated



Want: a world where everyone is not treated



IPW: Everything together

General rule: Each individual should be reweighted by

$$\text{Weight} = \frac{1}{\text{Prob of their obs treatment status given their age}}$$

- Propensity Score: $\pi_i = Pr(A_i = 1 \mid L_i)$
- Let N be the number of observations in the hypothetical world (which is the same as the number of actual observations)
- Hypothetical world where everyone is treated

$$\mathrm{E}(Y^{a=1}) = \frac{1}{N} \sum_{i:A_i=1} \frac{1}{\pi_i} \times Y_i$$

- Hypothetical world where everyone is not treated

$$\mathrm{E}(Y^{a=0}) = \frac{1}{N} \sum_{i:A_i=0} \frac{1}{1-\pi_i} \times Y_i$$

- $ATE =$ $\underbrace{\frac{1}{N} \sum_{i:A_i=1} \frac{1}{\pi_i} \times Y_i}$ $-$ $\underbrace{\frac{1}{N} \sum_{i:A_i=0} \frac{1}{1-\pi_i} \times Y_i}$

IPW: with data

Model the treatment assignment

$$\hat{\pi}_i = \hat{\mathrm{P}}(A = 1 \mid L) = \mathrm{logit}^{-1}\left(\hat{\alpha} + \hat{\gamma} L\right)$$

Estimate by inverse probability weighting

$$\hat{\mathrm{E}}(Y^{a=1}) = \frac{1}{N} \sum_{i:A_i=1} \frac{Y_i}{\hat{\pi}_i}$$

$$\hat{\mathrm{E}}(Y^{a=0}) = \frac{1}{N} \sum_{i:A_i=0} \frac{Y_i}{1 - \hat{\pi}_i}$$

Learning goals for today

At the end of class, you will be able to

- Estimate average causal effects via g-formula with a statistical model for the outcome and treatment
- Reason about the bias variance tradeoff when selecting a statistical model